The Averaged Periodogram: Welch's Method

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Key Concepts

1) The averaged periodogram is motivated by the poor variance properties of the periodogram.

2) Multiple nearly independent periodograms are averaged to reduce variance.

3) Welch's method divides the data into segments and averages the periodogram for each segment. The segments can overlap.

    a) The variance of the periodogram is reduced by the number of segments if the segments are independent.

b) Bias is due to the window associated with truncation and increases as the segment lengths decrease.

c) Windows are used to trade resolution (main lobe width) for dynamic range (side lobe height).

d) For a given window and fixed data length, increasing the number of segments reduces variance but increases bias.

Video Screenshots

Power Spectrum

$r_{xx}[k] \xleftrightarrow{\text{DTFT}} S_{xx}(\omega)$

Periodogram estimates $S_{xx}(\omega)$ from $x[n]$, $n = 0, 1, \cdots L-1$

$$x[n] \xleftrightarrow{\text{DFT}_L} X[k] \qquad \sim \frac{1}{L} |X[k]|^2$$

Asymptotically ($L \to \infty$) unbiased

$$E\{\hat{S}_{xx}^{per}(\omega)\} \to S_{xx}(\omega)$$

Variance behaves as $S_{xx}^2(\omega)$

$$Var\{\hat{S}_{xx}^{per}(\omega)\} \to S_{xx}^2(\omega)$$

Average multiple periodograms to reduce variance

Averaged Periodogram (Welch's method)

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- average K independent random variables: reduce variance by factor of K

- Divide $x[m]$, $m=0,1,\ldots Q-1$ into (possibly) overlapping segments of length L. r^{th} segment:

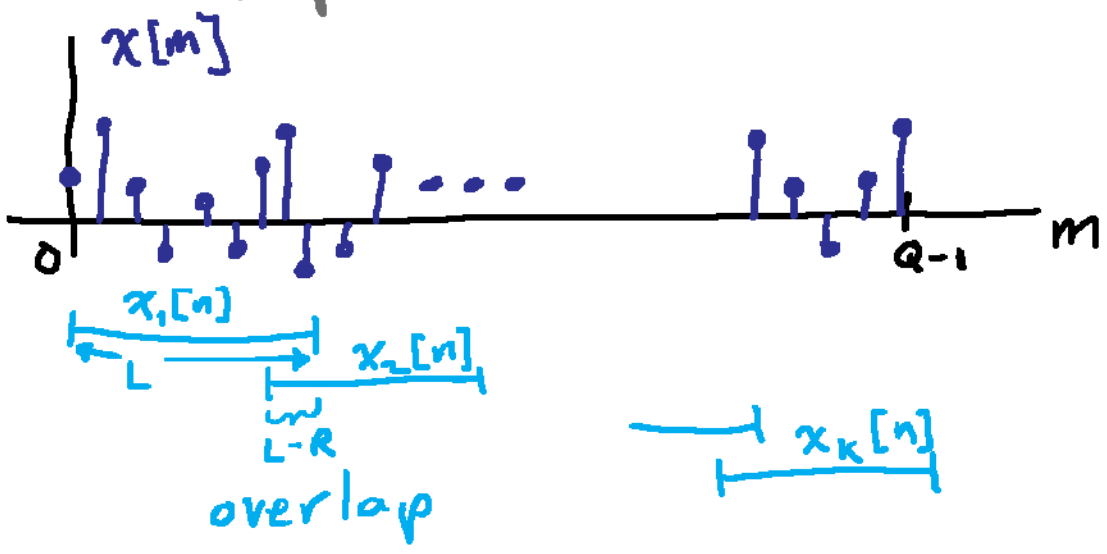
$$x_r[n] = x[rR+n]\,w[n], \quad n=0,1,\ldots L-1, \quad r=1,2,\ldots,K$$

optional window

- Average periodograms from each segment

$$I_r(\omega) = \frac{1}{2\pi \sum_{n=0}^{L-1} w^2[n]} \left| \sum_{n=0}^{L-1} x_r[n] e^{-j\omega n} \right|^2$$

$$\hat{S}_{xx}^{av}(\omega) = \frac{1}{K} \sum_{k=1}^{K} I_r(\omega)$$



Performance

3:44

Variance: $\mathrm{Var}\{\hat{S}_{xx}^{per}(\omega)\} \geq \mathrm{Var}\{\hat{S}_{xx}^{av}(\omega)\} \geq \frac{1}{K}\mathrm{Var}\{\hat{S}_{xx}^{per}(\omega)\}$

$1/K$ reduction if $I_r(\omega)$ are independent

Bias:

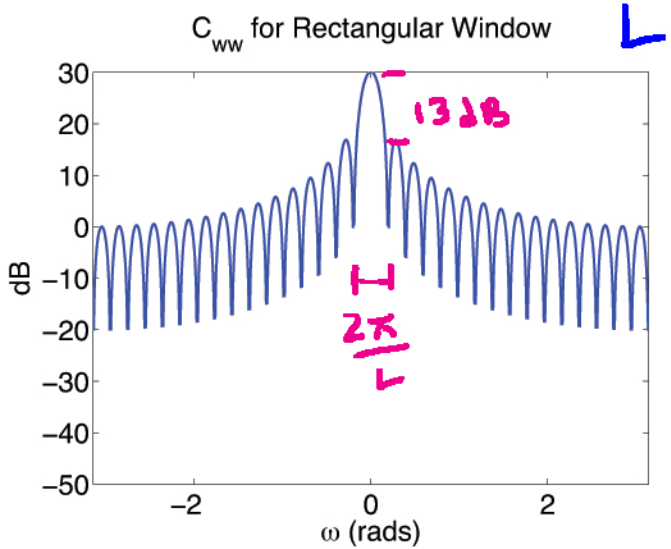
$$E\{\hat{S}_{xx}^{av}(\omega)\} = \frac{1}{2\pi LF}\int_{-\pi}^{\pi} S_{xx}(\theta)\, C_{ww}(e^{j(\omega-\theta)})\, d\theta$$

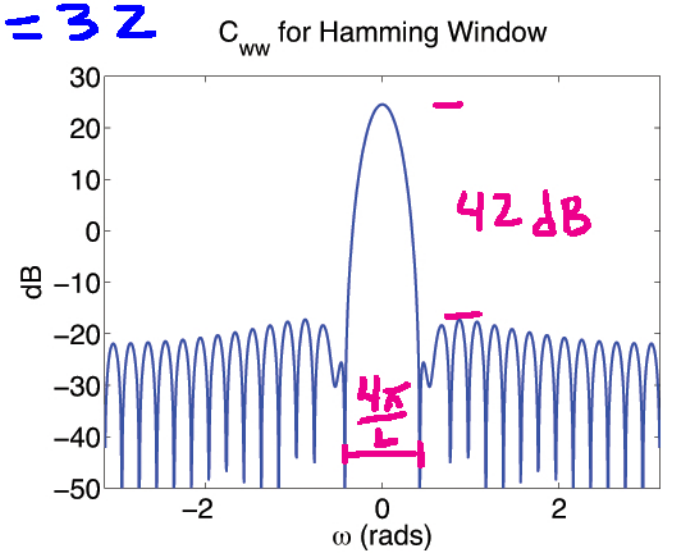
where $C_{ww}(e^{j\omega}) = |W(e^{j\omega})|^2$ $\quad w[n] \xleftrightarrow{DTFT} W(e^{j\omega})$

$$F = \frac{1}{L}\sum_{n=0}^{L-1} w^2[n]$$

Asymptotically ($L \to \infty$) unbiased

Resolution ↑ as L ↑





Tradeoffs between bias and variance

$$Q \geq (k-1)R + L$$

No overlap ⇒ maximum independence $(R = L)$

$$KL \leq Q$$

variance reduction

bias reduces as $L \uparrow$

Overlap $(R < L)$ allows both K and L to increase,
but reduces independence of segments (window dependent)

Windows manage bias

trade resolution (mainlobe width) for dynamic range (sidelobe height)

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